Food Delivery Time Analysis & Dashboard Project

Project Overview

This project focuses on analyzing food delivery operations using Python (Pandas) and Power BI. The objective was to identify factors affecting delivery time, clean and transform raw data, perform exploratory data analysis (EDA), and build an interactive dashboard that provides actionable business insights.

Objectives

- Clean and preprocess raw delivery data.
- Analyze delivery performance and delivery time patterns.
- Identify factors impacting delivery delays.
- Build an interactive Power BI dashboard.
- Provide recommendations to reduce average delivery time.

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI

Data Cleaning & Preprocessing

The following data preparation steps were performed using Pandas:

1. Data Inspection

- Checked dataset shape and structure.
- Reviewed data types.
- Identified missing values.
- Detected duplicate records.

2. Data Cleaning

- Removed extra spaces from text columns.
- Standardized categorical values.
- Cleaned Weather Conditions column.

- Replaced "unknown" values with NULL (NaN).
- Removed invalid or inconsistent records.

3. Date & Time Processing

- Converted Order Date into datetime format.
- Converted order and pickup times into proper time format.
- Renamed columns for consistency and readability.

4. Missing Value Handling

- Removed records with critical missing information.
- Dropped rows where order time was unavailable.

5. Export

- Saved the cleaned dataset for further analysis and dashboard creation.

Exploratory Data Analysis (EDA)

Several analyses were conducted to understand delivery performance:

Delivery Time Distribution

- Analyzed overall delivery time patterns.
- Identified delivery delays and outliers.

Traffic Impact Analysis

- Compared delivery time across different traffic densities.
- High traffic areas showed significantly longer delivery times.

Weather Impact Analysis

- Studied the effect of weather conditions on delivery performance.
- Adverse weather conditions increased delivery duration.

City-wise Analysis

- Compared order volumes across cities.
- Identified high-demand locations.

Correlation Analysis

- Generated correlation heatmap.
- Evaluated relationships among numerical variables.

Dashboard Features

The Power BI dashboard includes:

KPI Cards

- Total Orders
- Average Delivery Time
- Average Delivery Rating
- Total Delivery Partners

Visualizations

- Delivery Time by Traffic Density
- Delivery Time by Weather Condition
- City-wise Order Distribution
- Delivery Performance Trends
- Correlation Analysis

Interactive Filters

- City
- Weather Condition
- Traffic Density
- Vehicle Type
- Order Type

Key Insights

1. High traffic density significantly increases delivery time.
2. Poor weather conditions contribute to delivery delays.
3. Certain cities experience higher order volumes and operational pressure.
4. Delivery performance varies across vehicle types and locations.
5. Delivery partner ratings show a relationship with delivery efficiency.

Business Recommendations to Reduce Average Delivery Time

1. Traffic-Aware Order Assignment

Assign orders based on real-time traffic conditions and rider proximity.

Expected Impact: 10–15% reduction in delivery time.

2. Dynamic Route Optimization

Use GPS and route optimization algorithms to select the fastest routes.

Expected Impact: 8–12% improvement.

3. Increase Delivery Fleet During Peak Hours

Deploy additional delivery partners in high-demand areas.

Expected Impact: Reduced order backlog and waiting time.

4. Weather-Based Delivery Planning

Allocate additional riders and buffer time during adverse weather conditions.

Expected Impact: Improved service consistency.

5. Focus on High-Demand Cities

Create city-specific operational strategies for locations with high order volume.

6. Improve Rider Training

Train delivery partners on route planning and efficient delivery practices.

7. Establish Micro-Hubs

Create small distribution centers near high-order-density areas.

Expected Impact: Significant reduction in travel distance and delivery time.

Conclusion

This project demonstrates the complete data analytics workflow—from data cleaning and preprocessing using Pandas to dashboard development in Power BI. The analysis identified traffic density, weather conditions, and city-level demand as major factors influencing delivery time. By implementing the recommended operational improvements, businesses can reduce average delivery time, improve customer satisfaction, and increase overall delivery efficiency.